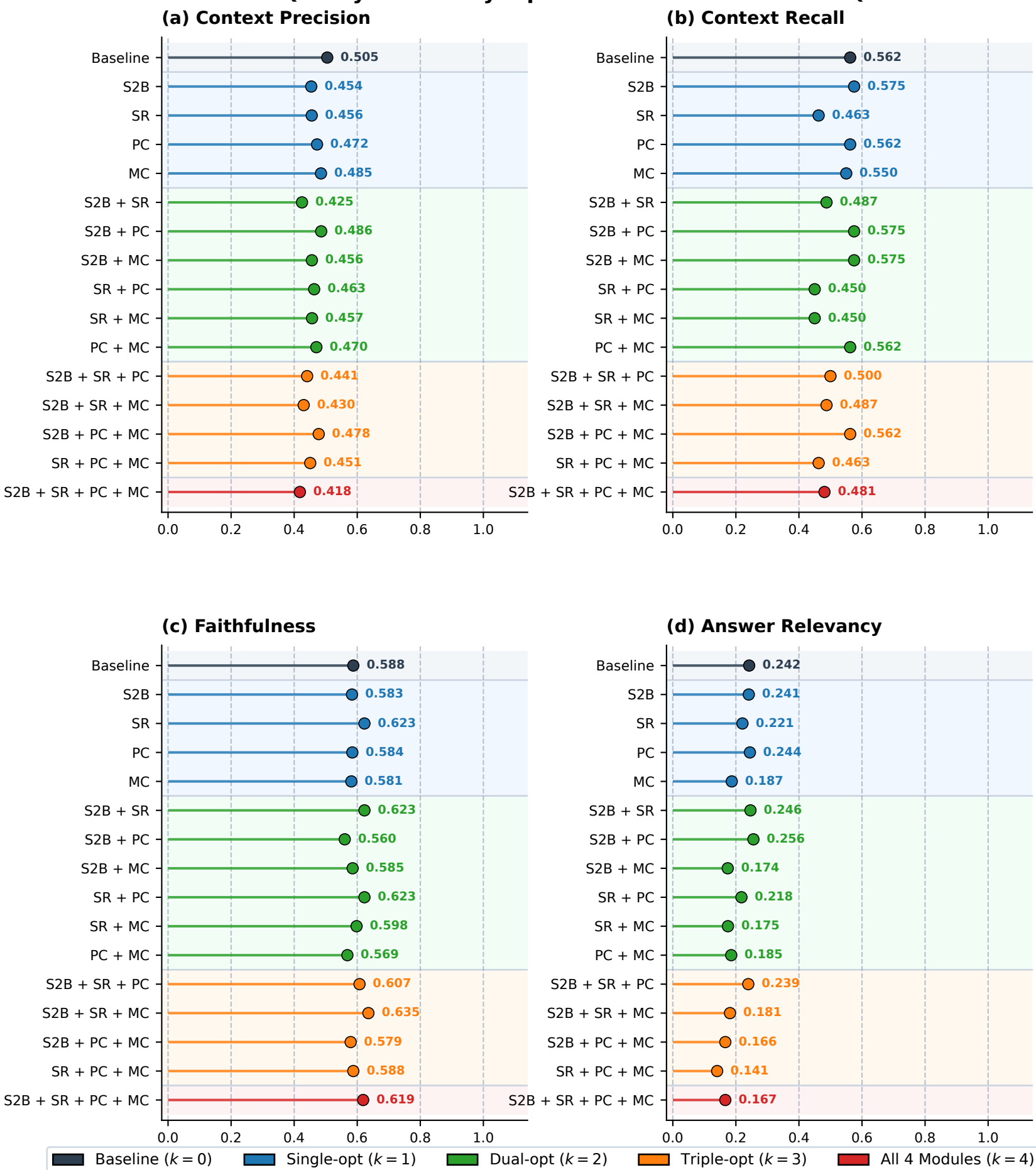
RAGAS Quality Metrics by Pipeline Architecture — MuSiQue
(a) Context Precision
Baseline 0.505
S2B 0.454
SR 0.456
PC 0.472
MC 0.485
S2B + SR 0.425
S2B + PC 0.486
S2B + MC 0.456
SR + PC 0.463
SR + MC 0.457
PC + MC 0.470
S2B + SR + PC 0.441
S2B + SR + MC 0.430
S2B + PC + MC 0.478
SR + PC + MC 0.451
S2B + SR + PC + MC 0.418
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 0.562
S2B 0.575
SR 0.463
PC 0.562
MC 0.550
S2B + SR 0.487
S2B + PC 0.575
S2B + MC 0.575
SR + PC 0.450
SR + MC 0.450
PC + MC 0.562
S2B + SR + PC 0.500
S2B + SR + MC 0.487
S2B + PC + MC 0.562
SR + PC + MC 0.463
S2B + SR + PC + MC 0.481
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.588
S2B 0.583
SR 0.623
PC 0.584
MC 0.581
S2B + SR 0.623
S2B + PC 0.560
S2B + MC 0.585
SR + PC 0.623
SR + MC 0.598
PC + MC 0.569
S2B + SR + PC 0.607
S2B + SR + MC 0.635
S2B + PC + MC 0.579
SR + PC + MC 0.588
S2B + SR + PC + MC 0.619
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.242
S2B 0.241
SR 0.221
PC 0.244
MC 0.187
S2B + SR 0.246
S2B + PC 0.256
S2B + MC 0.174
SR + PC 0.218
SR + MC 0.175
PC + MC 0.185
S2B + SR + PC 0.239
S2B + SR + MC 0.181
S2B + PC + MC 0.166
SR + PC + MC 0.141
S2B + SR + PC + MC 0.167
0.0 0.2 0.4 0.6 0.8 1.0
Baseline ($k = 0$)
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
All 4 Modules ($k = 4$)